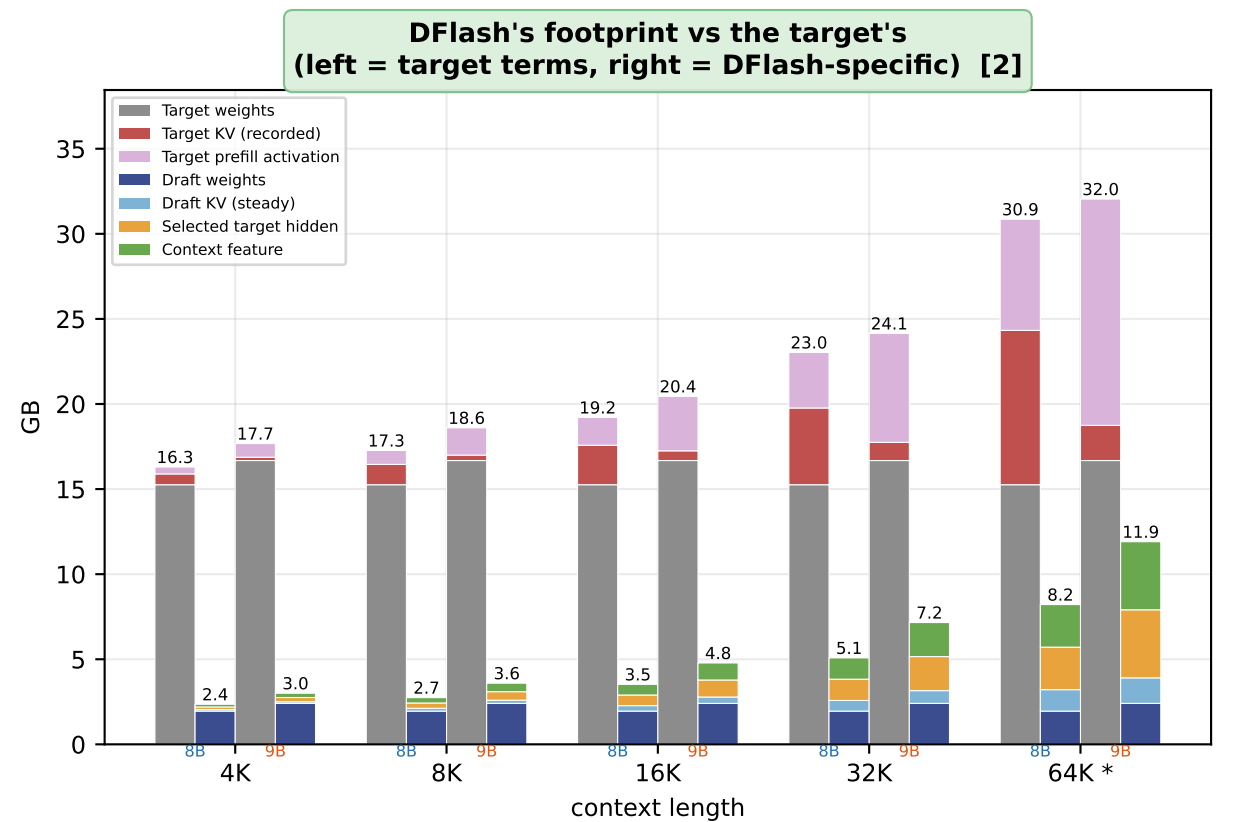
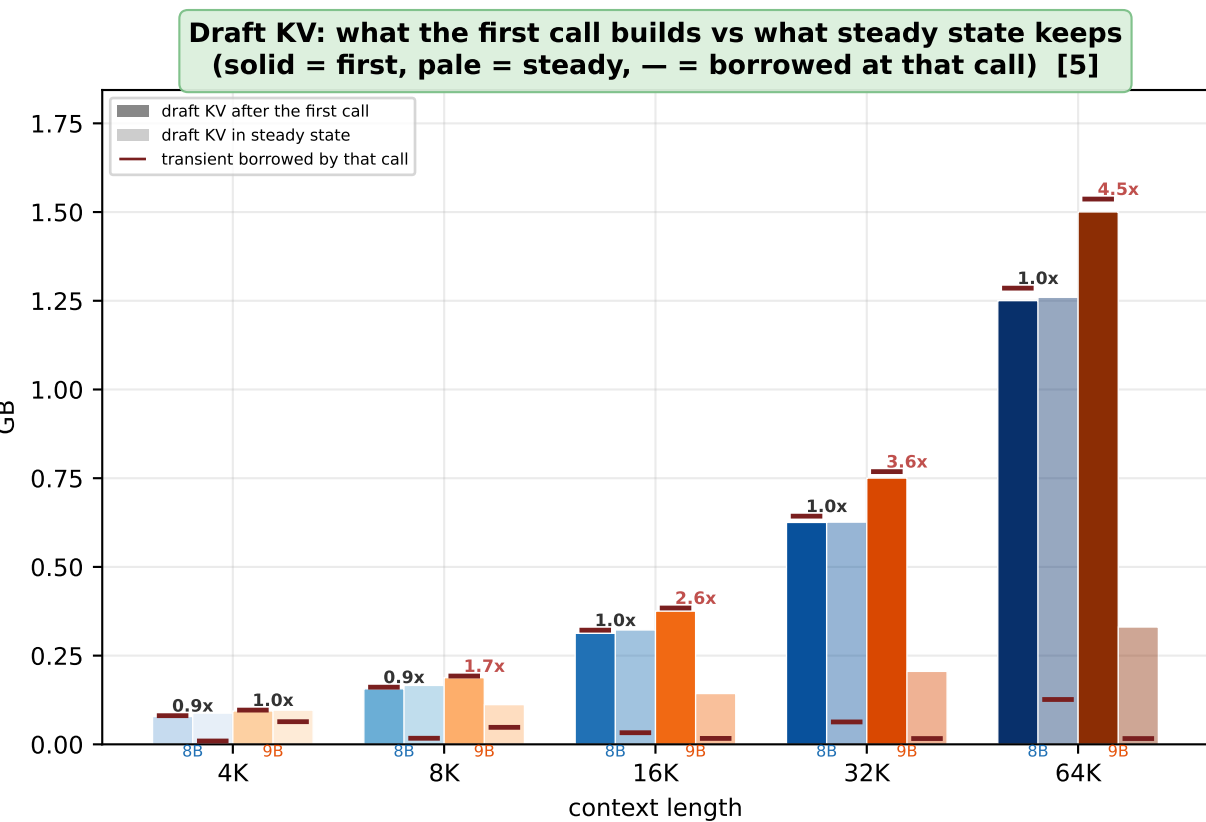
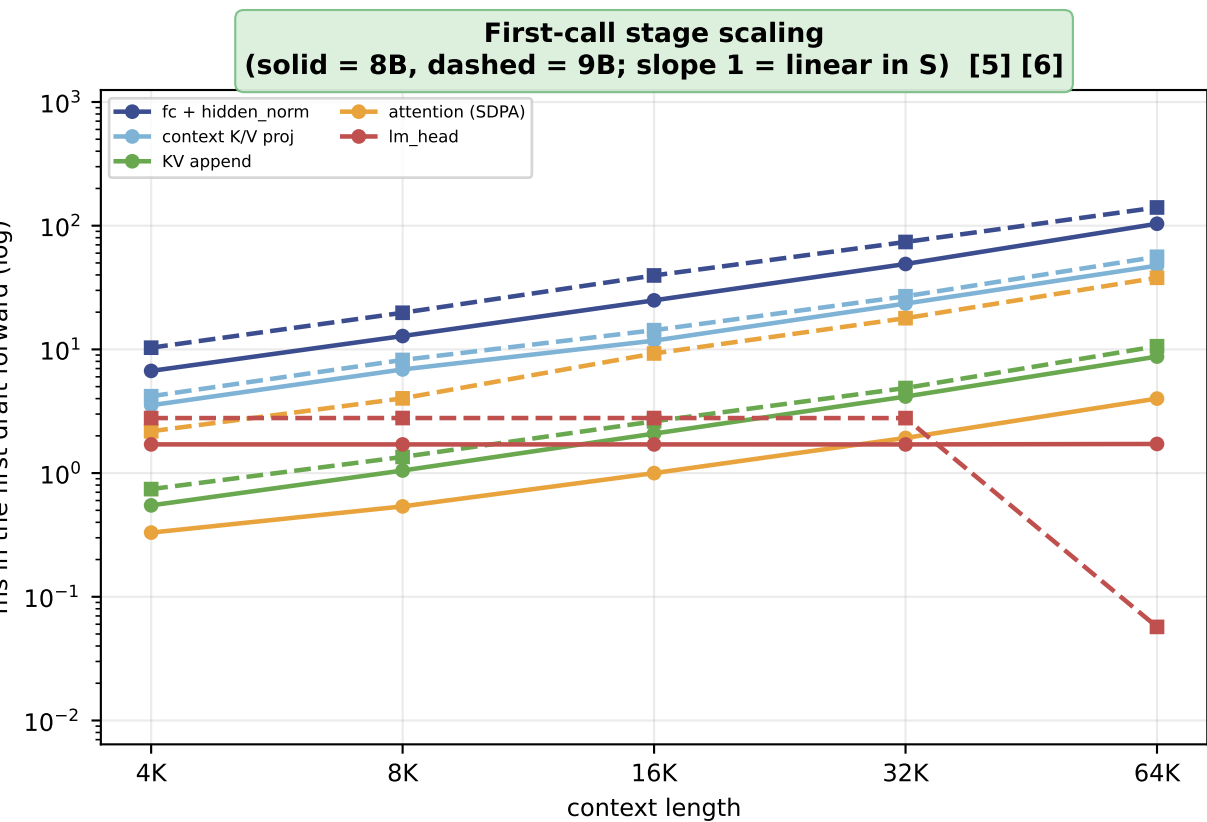
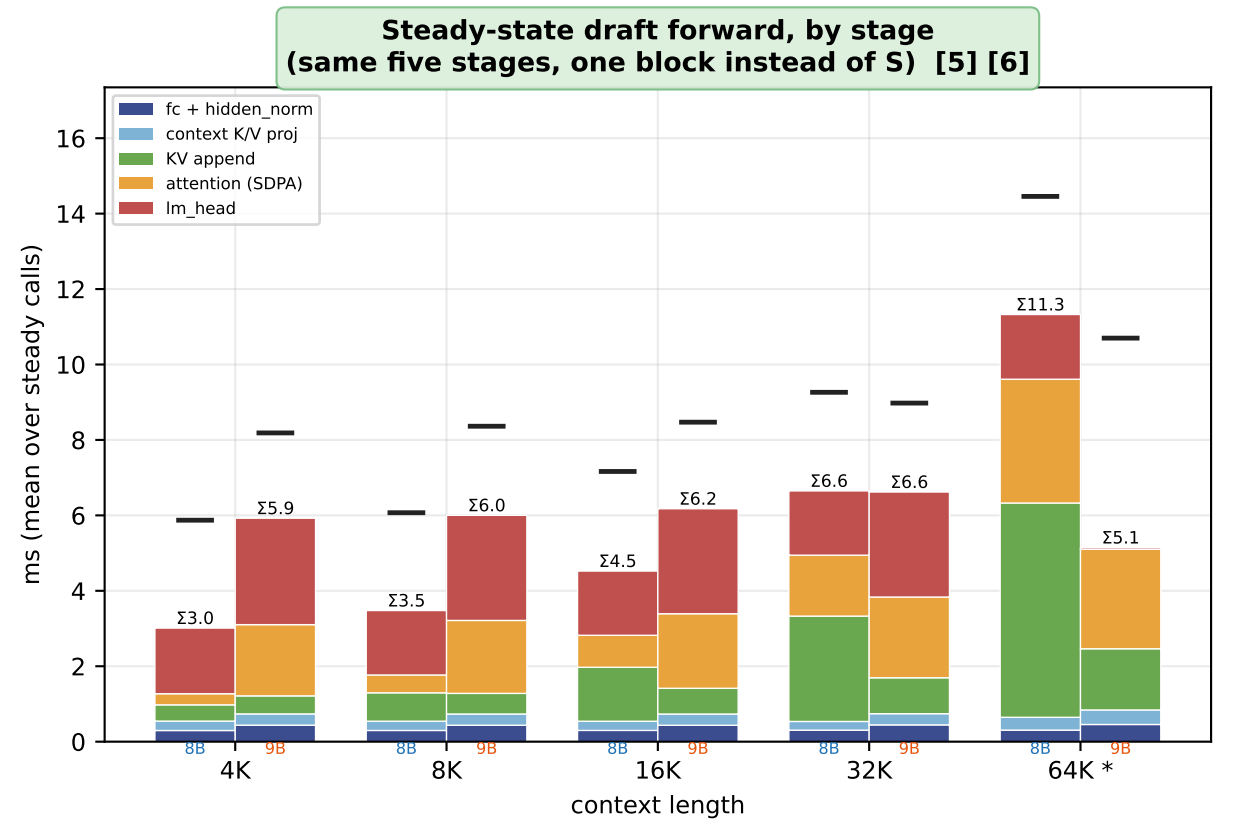
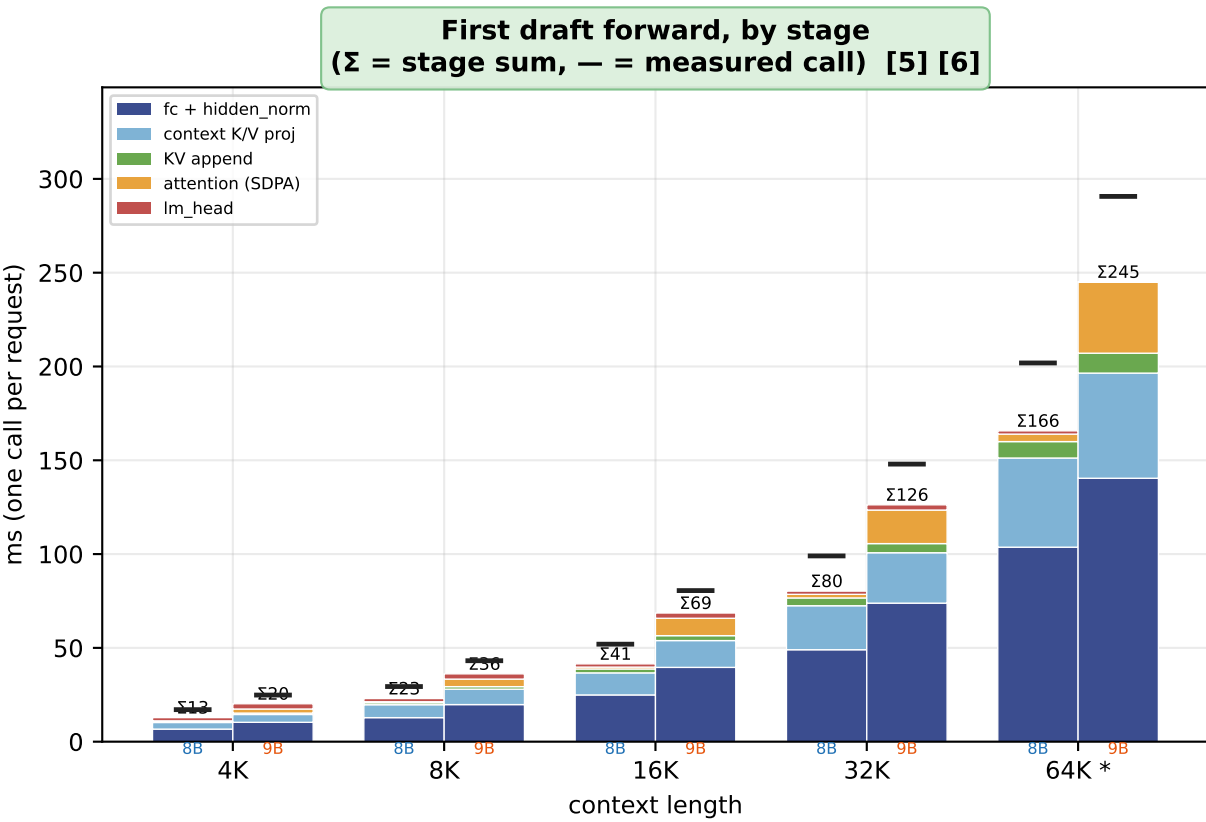
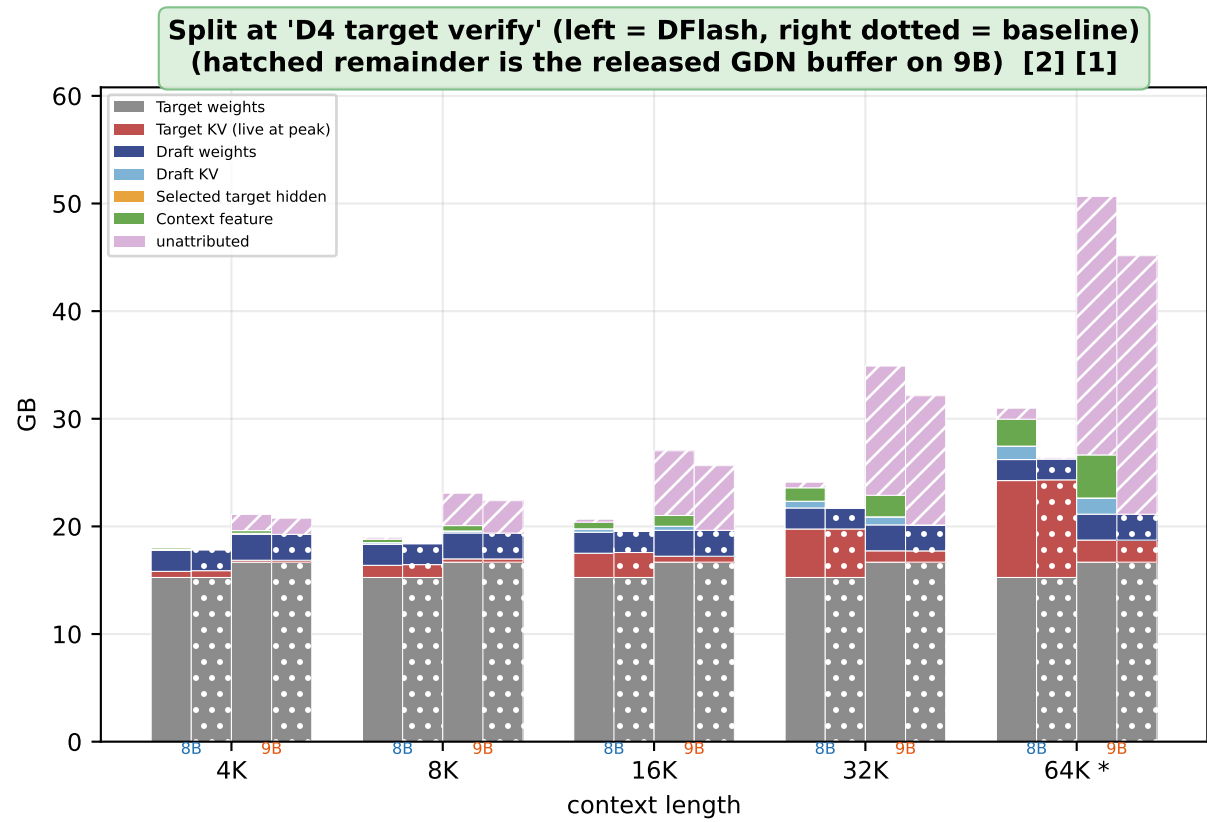
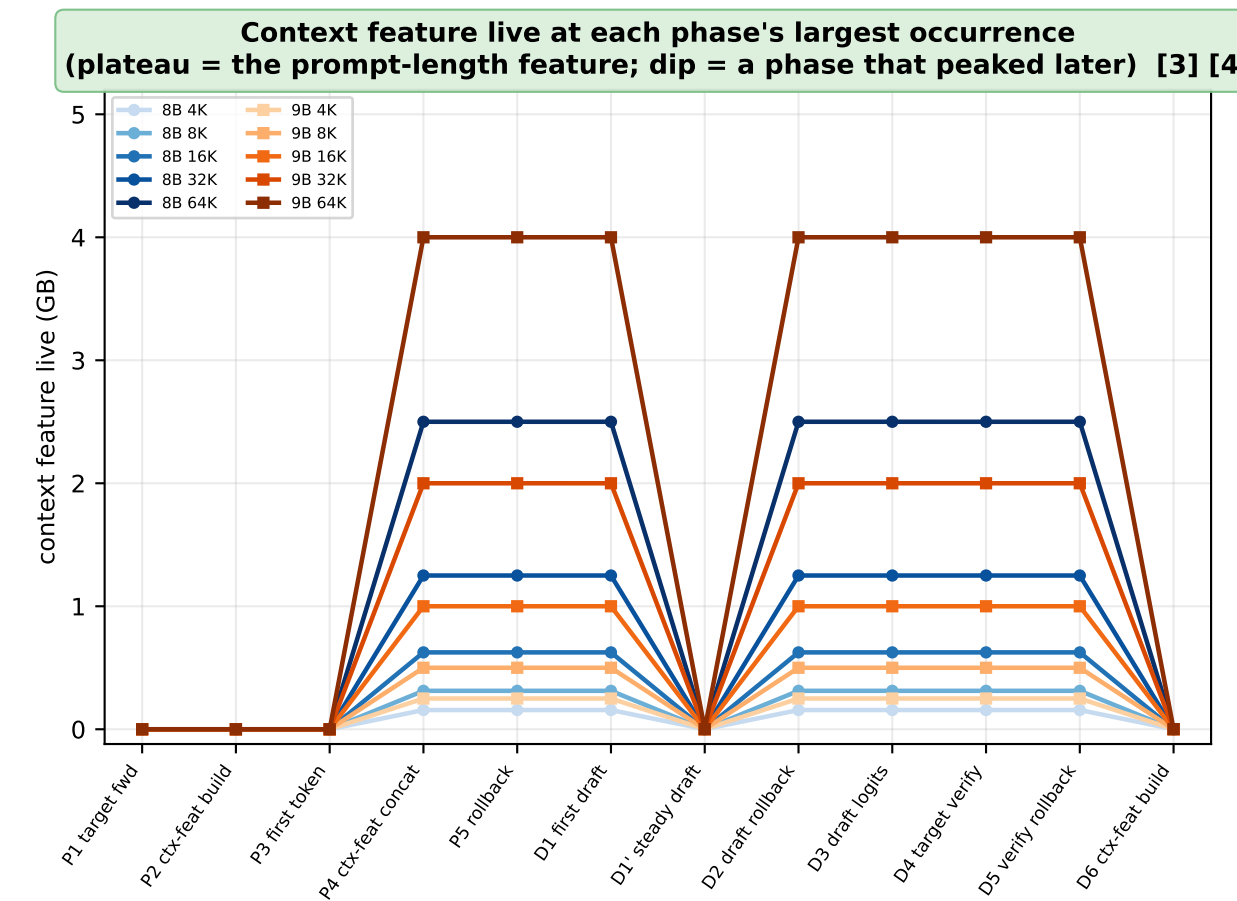
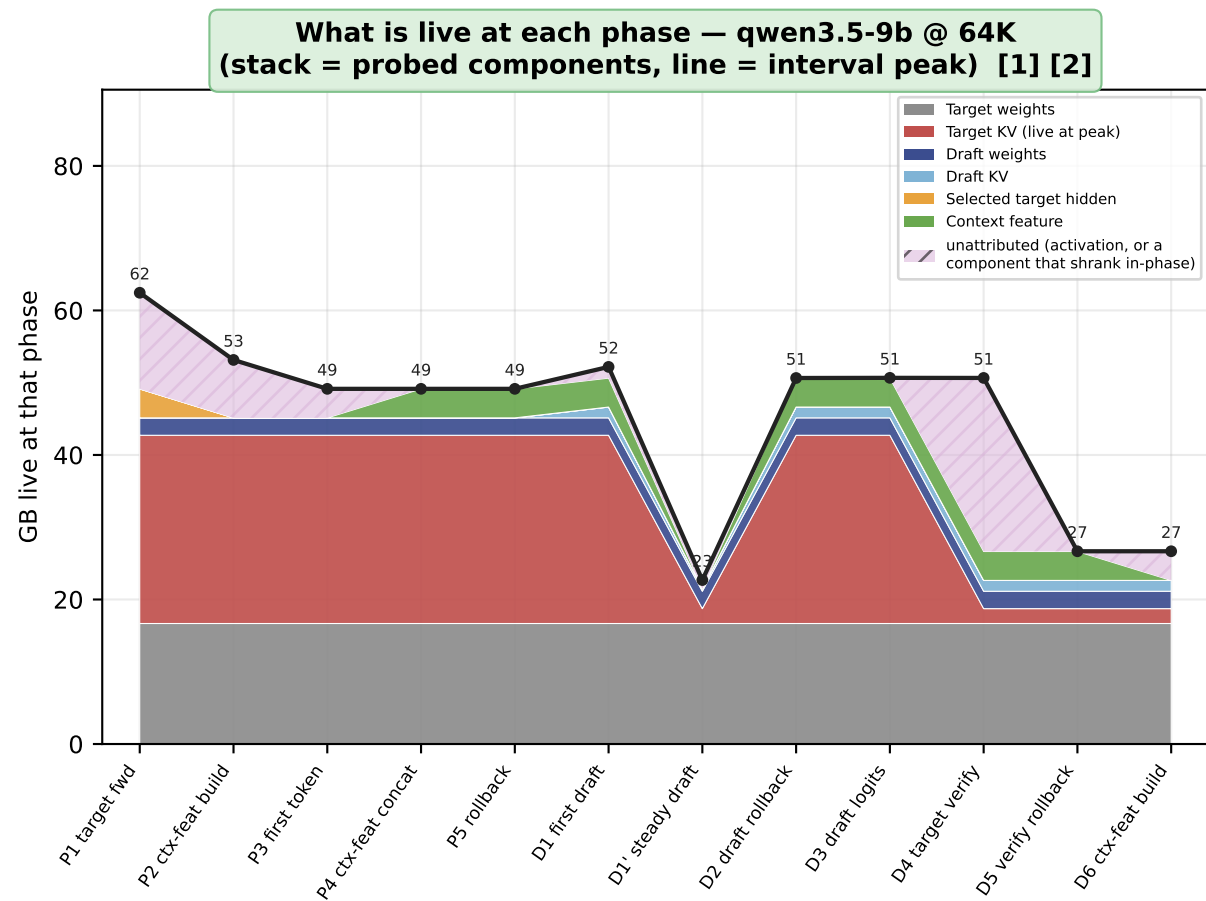
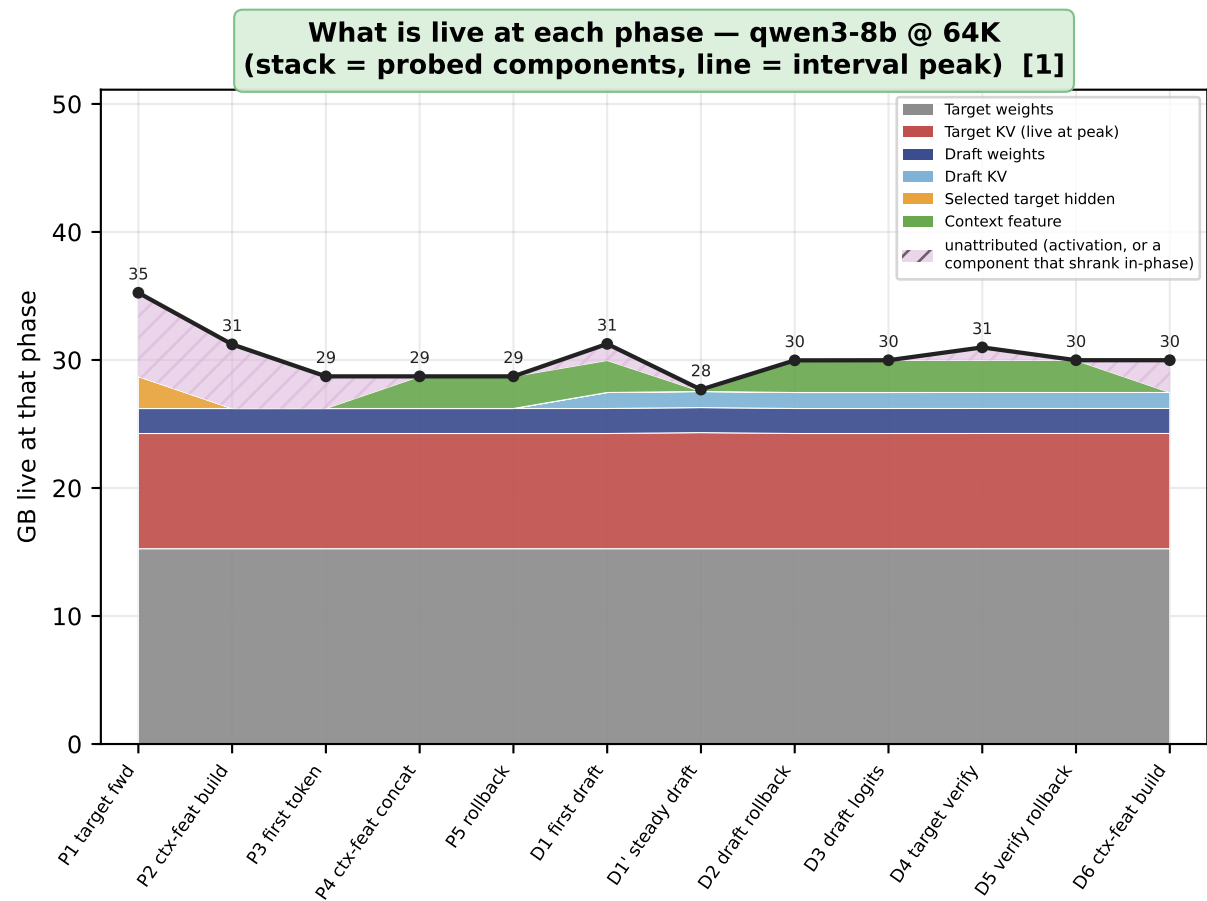


**DFlash phase-local memory and draft-stage profile — LongBench-E, hidden_states=selective
Qwen3-8B vs Qwen3.5-9B @ 4K / 8K / 16K / 32K / 64K · phases in execution order · block=16 · gamma=15**



qwen3-8b · 4K qwen3-8b · 8K qwen3-8b · 16K qwen3-8b · 32K qwen3-8b · 64K qwen3.5-9b · 4K qwen3.5-9b · 8K qwen3.5-9b · 16K qwen3.5-9b · 32K qwen3.5-9b · 64K

Every panel here is new — the pre-instrumentation records carried none of these fields, so there is no v1 figure to compare against. The notes say what each quantity is and where it can mislead

- [1] ADDED Each phase records the live component split probed at the *close* of its largest occurrence. A component that shrinks during the phase is therefore read at its post-shrink size, and the difference lands in 'unattributed'. That is exactly what happens to Qwen3.5-9B at 'D4 target verify': the GDN prefill buffer is live when the interval opens and gone when it closes, so target KV reads 2.05 GB against a 24 GB unattributed remainder. Read the two together at that phase, not separately.
- [2] ADDED Qwen3.5-9B's target KV drops from 26.05 GB to 2.05 GB at 64K on the first decode forward: 24 of its 32 layers are gated delta-rule, and their cache holds a sequence-length prefill buffer that is released once decoding starts. It is a target-architecture cost -- the same collapse is in the baseline's phase table, which never calls the drafter -- and must not be read as DFlash overhead.
- [3] ADDED The context feature is built at prefill over the whole prompt and read once, by the first draft forward. It then stays allocated through the draft rollback, the draft logits, the target verify and the verify rollback, until the next context-feature build replaces it -- 2.50 GB on 8B and 4.00 GB on 9B at 64K, held across phases that do not use it. Inference semantics are unchanged here; the panel records the lifetime rather than shortening it.
- [4] ADDED A phase reports its single largest occurrence, and different phases peak in different decode iterations, so a curve across phases is an envelope over the request rather than one timeline. The prompt-length context feature plateau (P4 through D5) is the first decode iteration; the dip at D1' is real and means the steady-state draft forward peaked later, when only the block-sized feature was live -- not that the feature was freed and rebuilt between D1 and D2.
- [5] ADDED first_draft_stage_s / mean_steady_*_s, new fields. A draft forward is timed in five stages with CUDA events: fc+hidden_norm over the injected feature, the context K/V projection, the cache append, the drafter's own SDPA, and the target's lm_head. The first call runs them over the whole context, every later call over one block.
- [6] REVISED 9B/64K is sharded over more than one card (starred on the x axis). Per-stage draft timings are recorded with CUDA events on the drafter's stream, so a stage whose weights live on the other card -- lm head, in practice -- is timed as the launch, not the work: it reads as ~0.05 ms against ~2.8 ms on a single card. Read the stage split of a starred column qualitatively only, and never subtract it from a single-GPU column.